

Machine Learning - MIRI Master (Final quiz - June 8, 2015)

Tick **clearly** the claims that you think are **true** - deliver just **this sheet** back

Name:

1. Complexity control and all that.

- ☐ If we keep complexity low, we do not need to care about training error
- ☐ Training error is always smaller (or equal) than test error
- ☐ Supplying more training data reduces the chances to obtain an overfitted model
- ☐ Regularization penalizes models that are either simpler or more complex than needed
- ☐ Training error is enough to perform model selection
- ☐ The VC dimension for a two-class classifier penalizes training sample size
- ☐ The VC dimension for a two-class classifier is the maximum number of linear separations that the classifier can perform
- ☐ In order to check that the VC dimension is (at least) some integer k , we just need to find k points that can be shattered
- ☐ Checking that the VC dimension is infinite requires an infinite number of checks
- ☐ A two-class classifier with infinite VC dimension must have an infinite (or very large) number of parameters

2. Bayesian classifiers.

- ☐ The Bayes formula converts prior distributions into posterior distributions
- ☐ The Bayes formula is of theoretical importance, but can never be used in practice
- ☐ The numerator in Bayes formula is enough to perform classification
- ☐ The Bayes classifier is the best possible classifier when the prior and posterior distributions are known
- ☐ For normally distributed classes, Bayesian classifiers turn out to be quadratic discriminant functions (QDA)
- ☐ For normally distributed classes, statistical independence among all variables yields linear discriminant functions (LDA)
- ☐ For normally distributed classes, Bayesian classifiers are minimum-distance classifiers
- ☐ The Naive-Bayes classifier assumes statistical independence among all variables
- ☐ The kNN classifier can be explained as a Bayesian classifier
- ☐ The kNN classifier works better with more neighbours, although it is computationally more costly

3. Maximum Likelihood and GLMs.

- ☐ The likelihood of a sample is its density for a given choice of parameters
- ☐ The likelihood of a sample is a function of the sample
- ☐ Logistic regression is a generative linear classifier
- ☐ Logistic regression assumes normally distributed classes
- ☐ In a Generalized Linear Model, the prediction is the logistic function applied to a linear function of the predictors
- ☐ The solution for Logistic regression can be found analytically by minimizing the log-likelihood
- ☐ In Poisson regression, we are interested in predicting integer outcomes, which are equally likely
- ☐ In a Generalized Linear Model, we always find the logistic function in one way or another (so it is called the link function)
- ☐ In statistics, bias and variance are opposite concepts: increasing one must decrease the other

- ❑ Variance always decreases with increasing sample size; however, bias can increase or stay the same

4. Regression, linear and non-linear.

- ❑ The regression function is the best possible model in regression, and achieves zero error on the training data
- ❑ The risk is equal to the sum of the (squared) bias, the variance and the noise variance
- ❑ The theoretical MSE does not depend on the regression function
- ❑ Models that are “more complex than needed” will tend to have a large bias and large variance
- ❑ Models that are “less complex than needed” will tend to have a small bias and small variance
- ❑ A linear combination of non-linear (fixed) functions of the inputs make a linear model
- ❑ Ridge regression adds a penalty term to a linear model such that the new model is non-linear
- ❑ Non-linear functions of the data can be estimated by using linear fitting techniques
- ❑ Both RBFs and MLPs create non-linear models by learning adaptive regressors (regressors with parameters)
- ❑ Regularization allows the specification of models that are more complex than needed; it also helps numerically

5. Kernels and SVMs.

- ❑ The kernel function defines kernel matrices whose elements are always positive
- ❑ The kernel function defines kernel matrices whose elements are always non-negative
- ❑ The product of two kernel functions may not be a kernel; however, the sum of two kernel functions is always a kernel
- ❑ By choosing a valid kernel, we get an inner product in a Hilbert space
- ❑ In order to kernelize a learning algorithm, this must be a supervised one, like the SVM
- ❑ The cost parameter (C) in a SVM acts as a regularizer of the solution
- ❑ Increasing the margin in a SVM leads to larger VC dimension
- ❑ Increasing the margin in a SVM leads to greater chances to separate the data
- ❑ Increasing the value of C in a SVM, the margin of the solution may increase
- ❑ Increasing the value of C in a SVM, the number of training errors may increase

6. Miscellaneous.

- ❑ The k-means algorithm converges to a global optimum if the number of iterations goes to infinity
- ❑ A Gaussian mixture model assumes normality of the training data
- ❑ The k-means algorithm is used to fine-tune a Gaussian mixture model after the latter has converged
- ❑ The backpropagation algorithm computes the partial derivatives of the error function with respect to the network weights
- ❑ The backpropagation algorithm must be coupled with an optimization method (update rule) to make it a learning algorithm for a MLP
- ❑ A MLP requires the specification of the number of hidden neurons; this is best done by trial-and-error, monitoring the fitting error of the network
- ❑ RBF neural networks are a particular case of MLP networks
- ❑ In a RBF neural network, regularization does not make sense, because it is based on Euclidean distance
- ❑ Feature selection can never increase the practical performance of a learning method, it only reduces learning time
- ❑ Feature selection can be performed after feature extraction, using the extracted features as new variables for selection